



Incus & Its Legacy

A 5-Minute Crash Course on System Containers and the Incus Hypervisor

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Linux Containers (**lxc**)



1. The Concept: System Containers

Docker popularized **application containers** (one process, disposable).

Incus comes from the **LXC** heritage of **System Containers** (a complete Linux OS).

- **It looks, feels, and acts like a Virtual Machine:**
 - Runs a complete Linux distribution: Debian, Ubuntu, Alpine, Arch, Fedora
 - Boots with a real init system (`systemd` or `OpenRC`) as PID 1
 - You can SSH in, create users, install packages with `apt`, and run daemons
- **...But with bare-metal speed:**
 - **Boots in 200ms**, consumes **~25MB RAM** at idle
 - Zero hypervisor overhead: runs directly on host cgroups & namespaces
- **Security by default:**
 - Unprivileged user namespaces (`UID 0` inside $\neq$ root on host) + AppArmor



2. The Engine: Real Storage & KVM VMs

Incus is not just a container runner—it is a **unified infrastructure manager**:

- **First-Class Storage:**
 - Pluggable drivers for **ZFS, Btrfs, LVM, and Linstor (DRBD)**
 - Instant sub-second copy-on-write snapshots, clones, and block-level quotas
- **The Unified Hypervisor (Containers + VMs):**
 - Need Windows, FreeBSD, or a custom kernel? Incus manages **KVM VMs** side-by-side
 - **One API and CLI for both:**

```
incus launch images:debian/trixie my-container    # System Container  
incus launch images:debian/trixie my-vm --vm      # KVM Virtual Machine
```



3. The History: How LXD Became Incus

2008: LXC (Linux Containers)

- Foundation of Linux containerization (Docker was originally an LXC wrapper!)



2015: LXD ("Lex-Dee")

- The container hypervisor under linuxcontainers.org



July 2023: The Canonical Takeover

- Canonical pulls LXD out of linuxcontainers.org, adds CLA, pushes Ubuntu Snaps



August 2023: Incus is Born

- Community fork led by original maintainers under linuxcontainers.org
- 100% open source (Apache 2.0), no CLA, distro-independent

- **Pronunciation:** LXD is "Lex-Dee"; Incus is "In-kus"



4. Modern Incus: Native OCI Application Containers

Incus modernized rapidly, solving the final puzzle piece:

- **The Reality:** Developers package applications as **Docker/OCI images**
- **Incus Native OCI:**
 - Incus can now pull directly from `docker.io`, `ghcr.io`, or any OCI registry
 - Runs application containers directly (natively, unprivileged, no nested Docker daemon)
- **The Trifecta in One REST API:**
 - i. **OCI Application Containers** (microservices)
 - ii. **LXC System Containers** (full OS userlands)
 - iii. **KVM Virtual Machines** (hardware virtualization)



5. The Bridge: Why `incus-compose`?

We have the ultimate Linux hypervisor. **So what was missing?**

- Developers have millions of `compose.yaml` files
- Nobody wants to write 20 imperative CLI commands to launch a multi-service stack

→ `incus-compose` is the drop-in bridge:

- Point it at your existing `compose.yaml` → `incus-compose up -d`
- Runs directly on Incus with official `compose-go` parsing
- Handles health checks, restart policies, and dependency gating via `ic-healthd`
- Instant ZFS/Linstor volume backups and zero-touch ingress

Now, let's look at the main presentation and see it in action!